(ix) Vector identities

$$(\vec{a}+\vec{b})^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a}\cdot\vec{b}$$

$$(\vec{a}-\vec{b})^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a}\cdot\vec{b}$$

$$(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b}) = |\vec{a}|^2 - |\vec{b}|^2$$

Proof

By property (iii)

$$(\vec{a}+\vec{b})^2 = (\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b}) = |\vec{a}|^2 + |\vec{b}|^2 + \vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{a} = |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a}\cdot\vec{b}$$

Similarly one can prove other results. ■

(x) Working rule to find scalar product of two vectors

Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$$\begin{aligned}\vec{a}\cdot\vec{b} &= (a_1\hat{i} + a_2\hat{j} + a_3\hat{k})\cdot(b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) \\ &= a_1b_1(\hat{i}\cdot\hat{i}) + a_1b_2(\hat{i}\cdot\hat{j}) + a_1b_3(\hat{i}\cdot\hat{k}) \\ &\quad + a_2b_1(\hat{j}\cdot\hat{i}) + a_2b_2(\hat{j}\cdot\hat{j}) + a_2b_3(\hat{j}\cdot\hat{k}) + a_3b_1(\hat{k}\cdot\hat{i}) + a_3b_2(\hat{k}\cdot\hat{j}) + a_3b_3(\hat{k}\cdot\hat{k}) \\ &= a_1b_1 + a_2b_2 + a_3b_3.\end{aligned}$$

Therefore, the scalar product of two vectors is equal to the sum of the products of their corresponding rectangular components. ■

(xi) If θ is the angle between the vectors $\vec{a}$ and $\vec{b}$ then $\theta = \cos^{-1}\left[\dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}|\,|\vec{b}|}\right]$.

(xii) For any two vectors $\vec{a}$ and $\vec{b}$, $|\vec{a}+\vec{b}| \le |\vec{a}| + |\vec{b}|$.

We know that if $\vec{a}$ and $\vec{b}$ are the two sides of a triangle then the sum $\vec{a}+\vec{b}$ represents the third side of the triangle. Therefore, by triangular property, $|\vec{a}+\vec{b}| \le |\vec{a}| + |\vec{b}|$ ■

(xiii) For any two vectors $\vec{a}$ and $\vec{b}$, $|\vec{a}\cdot\vec{b}| \le |\vec{a}||\vec{b}|$.

If one of them is zero vector then the equality holds. So, let us assume that both are non-zero vectors. We have

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$$

That is, $\dfrac{|\vec{a}\cdot\vec{b}|}{|\vec{a}||\vec{b}|} = |\cos\theta| \le 1$

$$\Rightarrow |\vec{a}\cdot\vec{b}| \le |\vec{a}||\vec{b}|.$$ ■

Example 8.11

Find $\vec{a}\cdot\vec{b}$ when

(i) $\vec{a} = \hat{i} - \hat{j} + 5\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{k}$

(ii) $\vec{a}$ and $\vec{b}$ represent the points (2, 3, – 1) and (– 1, 2, 3).

Solution

(i) $\vec{a}\cdot\vec{b} = (\hat{i} - \hat{j} + 5\hat{k})\cdot(3\hat{i} - 2\hat{k}) = (1)(3) + (-1)(0) + (5)(-2) = 3 - 10 = -7$

(ii) $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = -\hat{i} + 2\hat{j} + 3\hat{k}$

$\vec{a}\cdot\vec{b} = (2)(-1) + (3)(2) + (-1)(3) = -2 + 6 - 3 = 1.$

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